

Teacher Guide — primary / module-12- movement-of-sun

IKS Curriculum

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Overview

Movement of the Sun

Band: Primary (Grades 3–5) · **Sessions:** 10 × 30 min · **Sanskrit:** Sūrya-gati — Uttarāyana and Dakṣiṇāyana

Overview

Earth’s axial tilt produces an apparent annual path of the Sun along the ecliptic. Indian astronomy names the northward course *Uttarāyana* and southward course *Dakṣiṇāyana*, with 12 *Rāśi* (zodiacal divisions) and *saṁkrānti* moments when the Sun crosses between them. Sāyana (tropical) vs nirayana (sidereal) reference frames are introduced for senior students.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning (sunrise observation)
- §11.8 — Holistic, multidisciplinary (astronomy + calendar)

Key terms (this module)

- *Sūrya* — the Sun
- *Uttarāyana* — northward solar course (Dec 21 – June 21)
- *Dakṣiṇāyana* — southward solar course (June 21 – Dec 21)
- *saṁkrānti* — Sun’s crossing from one Rāśi to the next
- *rāśi* — 1/12th of the ecliptic; zodiacal sign
- *sāyana* — tropical zodiac (linked to seasons)
- *nirayana* — sidereal zodiac (linked to fixed stars)

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper
- (See [lesson-plan.md](#) for per-day material lists)

10-day map

- **Day 1** — The Sun's Yearly Path: Observe that sunrise direction changes; introduce the ecliptic.
- **Day 2** — Axial Tilt and the Seasons: Demonstrate with torch+globe why we have summer and winter.
- **Day 3** — Uttarāyaṇa and Dakṣiṇāyana: Identify the dates and direction of each.
- **Day 4** — Equinoxes — Viṣuva: Identify spring and autumn equinoxes (day = night).
- **Day 5** — Twelve Rāśis: Name the 12 rāśis in Sanskrit and pair with Greek zodiac names.
- **Day 6** — Saṁkrānti Moments: Identify Makara, Karka, and other saṁkrāntis on the calendar.
- **Day 7** — Sāyana vs Nirayana: Distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.
- **Day 8** — Precession of the Equinoxes: Explain why Indian and Western zodiac dates differ today.
- **Day 9** — Cross-Cultural Solar Calendars: Compare with Egyptian, Roman, Gregorian.
- **Day 10** — Final Share and Assessment: Project share + summative quiz.

Files in this pack

- [lesson-plan.md](#) — detailed 10-day arc
- [sources.md](#) — sources cited in this module
- [teacher-notes.md](#) — common misconceptions, differentiation
- [parent-guide.md](#) — what parents should know
- [days/day-NN-*.md](#) — 10 per-day lesson plans
- [quizzes/formative.md](#), [quizzes/summative.md](#)
- [activities/activity-01-*.md](#)
- [assessments/rubric.md](#), [assessments/project-brief.md](#)
- [student-workbook.md](#)
- [writing-prompts.md](#) (or [oral-recall.md](#) for Primary)
- [homework.md](#)
- [slides/deck.md](#)

Voice

Story-led, sensory, drawing-heavy; 8–12 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Movement of the Sun (Primary)

10-day arc, 30 min per session.

Day 1 — The Sun's Yearly Path

Objective: Observe that sunrise direction changes; introduce the ecliptic.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 2 — Axial Tilt and the Seasons

Objective: Demonstrate with torch+globe why we have summer and winter.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

Objective: Identify the dates and direction of each.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 4 — Equinoxes — Viṣuva

Objective: Identify spring and autumn equinoxes (day = night).

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 5 — Twelve Rāśis

Objective: Name the 12 rāśis in Sanskrit and pair with Greek zodiac names.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 6 — Saṁkrānti Moments

Objective: Identify Makara, Karka, and other saṁkrāntis on the calendar.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 7 — Sāyana vs Nirayana

Objective: Distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 8 — Precession of the Equinoxes

Objective: Explain why Indian and Western zodiac dates differ today.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 9 — Cross-Cultural Solar Calendars

Objective: Compare with Egyptian, Roman, Gregorian.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 10 — Final Share and Assessment

Objective: Project share + summative quiz.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Pacing notes

- Primary sessions are 30 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Movement of the Sun (Primary)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.

- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition's framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji's mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn't?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Movement of the Sun (Primary)

Primary and secondary sources

- Sūrya-siddhānta (c. 5th–10th c. CE), chapters on solar motion
- Āryabhaṭīya (499 CE), on precession
- Varāhamihira, *Bṛhat Saṃhitā* (6th c. CE)
- NCERT Physics Class XI, ch. 8 (Gravitation)
- NCERT Geography Class IX, ch. 3 (Motions of the Earth)

A note on citation discipline

Per the curriculum's style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Primary-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at `curriculum/_shared/glossary.md` contains the IAST + meaning for every Sanskrit term used.

- Other modules in this curriculum that connect: see the module README’s “Prerequisites” section.

Day 1 — The Sun’s Yearly Path

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Observe that sunrise direction changes; introduce the ecliptic.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)
- The Sun rises in different places through the year!

Stand at a window facing east. Mark where the Sun comes up TODAY. (Just outside the window — point with arms.)

In summer, the Sun rises a bit more to the LEFT. In winter, more to the RIGHT (in northern places).

Children draw a window with the Sun in 3 spots: summer (left), middle (spring/autumn), winter (right). One drawing.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — Axial Tilt and the Seasons

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Demonstrate with torch+globe why we have summer and winter.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Uttarāyaṇa* (IAST: Uttarāyaṇa; lit. “northward solar course (Dec 21 – June 21)”)
- Why do we have seasons?

Demo: hold a ball (Earth) tilted to one side. Shine a torch (Sun) at it. Move the ball around the torch. - Sometimes the TOP of the ball gets more light → summer. - Sometimes the BOTTOM gets more → other hemisphere's summer.

“*Earth is TILTED. That's why we have seasons!*” Children draw a tilted Earth with the Sun.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 3 — Uttarāyana and Dakṣiṇāyana

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Identify the dates and direction of each.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)
 - **Uttarāyana** (“Going North”): Dec to June — days get longer in the north.
 - **Dakṣiṇāyana** (“Going South”): June to Dec — days get shorter in the north.

Children stand with hands held high. Drift to the right (north). “*That's Uttarāyana!*” Drift to the left (south). “*That's Dakṣiṇāyana!*”

Each child draws an arrow on a calendar showing today's direction.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 4 — Equinoxes — Viṣuva

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Identify spring and autumn equinoxes (day = night).

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *saṁkrānti* (IAST: saṁkrānti; lit. “Sun's crossing from one Rāśi to the next”)
- Two special days each year: day and night are EXACTLY the same length!

These are called **equinoxes** (or *viṣuva*). - **Spring equinox** (around March 21): days getting longer. - **Autumn equinox** (around September 22): days getting shorter.

Children stand still with arms balanced equally. “*Day equals night!*”

Draw two suns on a year-circle, one in March, one in September. Label “equinox.”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 5 — Twelve Rāśis

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Name the 12 rāśis in Sanskrit and pair with Greek zodiac names.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *rāśi* (IAST: rāśi; lit. “1/12th of the ecliptic; zodiacal sign”)
- The Sun visits 12 special “houses” each year!

Names (sing-chant): “*Meṣa, Vṛṣabha, Mithuna, Karka, Siṃha, Kanyā, Tulā, Vṛścika, Dhanus, Makara, Kumbha, Mīna!*”

Each “house” has a Sanskrit name AND a Greek name: - Meṣa = Aries (ram!), Vṛṣabha = Taurus (bull!), Mithuna = Gemini (twins!)...

Children pick one rāśi to draw. Today’s question: “*What’s YOUR birth-month rāśi?*”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 6 — Saṃkrānti Moments

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Identify Makara, Karka, and other saṃkrāntis on the calendar.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today's idea (10 min)

- *sāyana* (IAST: sāyana; lit. “tropical zodiac (linked to seasons)”)
- When the Sun moves from one rāśi-house to the next, we call it *San̥krānti*.
- The most famous: *Makara San̥krānti* — when the Sun enters Makara (Capricorn). Around January 14 in many years!

Show a wheel with 12 houses. Children put a Sun-sticker on each rāśi, then circle the date when the Sun enters Makara.

“In Tamil Nadu, this is Pongal. In Punjab, Lohri. In many places, it's a happy festival!”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *nirayana* (IAST: nirayana; lit. “sidereal zodiac (linked to fixed stars)”)
- There are TWO ways to measure the Sun’s path:
 - **Sāyana** (modern science): starts at the Vernal Equinox (Mar 21).
 - **Nirayana** (Indian tradition): starts at a fixed star.

These two ways DRIFT apart slowly — about 1° every 70 years. Over many centuries, they’re now ~24 days apart.

Children draw two suns: one labelled “tradition” and one labelled “modern.” They’re at slightly different spots.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Precession of the Equinoxes

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Explain why Indian and Western zodiac dates differ today.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *Sūrya* (IAST: Sūrya; lit. "the Sun")
- The Earth wobbles slowly, like a spinning top.
- One wobble takes about 26,000 years!

Show a spinning top toy. Let it slow down — it wobbles. "*Earth does this too, just SUPER slowly.*"

Children draw Earth tilted, with a curved arrow showing the wobble. "*Astronomers from long ago noticed this!*"

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Cross-Cultural Solar Calendars

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Compare with Egyptian, Roman, Gregorian.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Uttarāyaṇa* (IAST: Uttarāyaṇa; lit. “northward solar course (Dec 21 – June 21)”)
- Many people use the Sun to make calendars!
- Egypt: 365 days, like ours.
- India: months named after the rāśis.
- Roman/modern: January, February, ...

Children write today's date in 3 ways: 1. Modern (Jan 25, 2026) 2. Tamil (rāśi-based) 3. Just by season (winter/late winter)

Discuss which they like.

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Share and Assessment

Module: Movement of the Sun (Primary) · **Time:** 30 min

Goal

Project share + summative quiz.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)
- Final day! We celebrate the Sun’s yearly journey.

Each child shows their year-wheel. Cheer for each.

Sing the 12 rāśi names together one more time.

Quiz with stickers — everyone gets a “Sun Watcher” sticker. “*Never look at the Sun! Use eyes-down only.*”

Activity (8 min)

A simplified version of *Sunrise Direction Tracking* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.

- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Formative Quiz — Movement of the Sun (Primary)

When: mid-module (after Day 5). **Duration:** 20 minutes. **Format:** mix of short answer, multiple choice, and reflection. Sticker self-assessment for Primary.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Movement of the Sun in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Uttarāyaṇa and Dakṣiṇāyana’?
(a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Movement of the Sun is purely a religious concept and has no scientific framing.*
5. Connect Movement of the Sun to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘Sūrya’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Movement of the Sun and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Movement of the Sun (Primary)

When: end of module (after Day 10). **Duration:** 30 minutes. **Total points:** 20.

Questions

Part A — Vocabulary (~5 points) A1. Define *Sūrya* in your own words. A2. Define *Uttarāyaṇa* in your own words. A3. Define *Dakṣiṇāyana* in your own words. A4. Define *saṁkrānti* in your own words. A5. Define *rāśi* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Movement of the Sun in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4.

Compare Movement of the Sun with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Movement of the Sun from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 12/20 indicates the student has met the module's learning objectives.

Homework

Homework — Movement of the Sun (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Sun's Yearly Path** · Draw about today's concept. (~5 min)
2. **Day 2 — Axial Tilt and the Seasons** · Draw about today's concept. (~5 min)
3. **Day 3 — Uttarāyana and Dakṣiṇāyana** · Draw about today's concept. (~5 min)
4. **Day 4 — Equinoxes — Viṣuva** · Draw about today's concept. (~5 min)
5. **Day 5 — Twelve Rāsis** · Draw about today's concept. (~5 min)
6. **Day 6 — Saṃkrānti Moments** · Draw about today's concept. (~5 min)
7. **Day 7 — Sāyana vs Nirayana** · Draw about today's concept. (~5 min)
8. **Day 8 — Precession of the Equinoxes** · Draw about today's concept. (~5 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Share and Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Activity 1 — Sunrise Direction Tracking

Module: Movement of the Sun · **Band:** Primary · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Movement of the Sun through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

compass app, classroom window, notebook

What students do

Over 5 days, students note the compass bearing of sunrise each morning (from the same window). They graph the bearing and predict how it will change in 3 months.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Personal Solar Calendar

Module: Movement of the Sun · **Band:** Primary

What you will do

Make a one-page solar calendar for the current year: mark the two solstices, two equinoxes, all 12 saṃkrānti dates, plus your own birthday with its rāśi.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a drawing / made object).
- **Written reflection** (oral share): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (3-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 1 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Movement of the Sun (Primary)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

- **Knows the new word(s)** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Joined in activities** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Asked good questions** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Drew / made something** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.